

NHL Trade Market Efficiency — Research Brief

Prepared to support a Master's paper titled "NHL Trade Market Efficiency Model: Identifying Systematic Biases in Asset Valuation." Organized by the eight research areas in the task. Peer-reviewed sources are flagged explicitly; industry analytics sources (blogs, conference presentations) are cited with author and venue. Where consensus exists vs. methodologies diverge, this is flagged.

1. Sports economics & market efficiency literature

1A. Seminal sports labor market efficiency papers

- **Scully, G. W. (1974).** "Pay and Performance in Major League Baseball." *American Economic Review* 64(6): 915–930. [JSTOR](#) Canonical marginal revenue product (MRP) framework: two-equation system (team revenue on win%; wins on player inputs) imputes MRP. [Ecu](#) Under the reserve clause, players earned only **10–20% of MRP** [Ecu](#) — evidence of monopsonistic exploitation. Spawned the sports MRP literature (Medoff 1976; Zimbalist 1992; Krautmann 1999).
- **Krautmann, A. (1999).** "What's Wrong with Scully-Estimates of a Player's MRP." *Economic Inquiry* 37(2): 369–381. Introduces the "free-market returns" method; [ResearchGate](#) argues Scully estimates are upwardly biased.
- **Humphreys, B. & Pyun, H. (2017).** "Monopsony Exploitation in Professional Sport: Evidence from MLB Position Players, 2000–2011." *Managerial and Decision Economics*. Estimates Monopsony Exploitation Ratios $\approx$ 0.89 rookies, 0.75 arb-eligible, 0.21 free agents [ResearchGate](#) — exploitation persists but concentrated on pre-FA players. Directly relevant to NHL ELC surplus-value arguments.
- **Kahn, L. M. (2000).** "The Sports Business as a Labor Market Laboratory." *Journal of Economic Perspectives* 14(3): 75–94. [American Economic Association](#) Standard framing citation: sports data richness (salary, output, contracts publicly observable) makes it a uniquely clean laboratory for testing monopsony, discrimination, Coase, and tournament theory.
- **Rosen, S. & Sanderson, A. (2001).** "Labour Markets in Professional Sports." *Economic Journal* 111(469): 47–68. Comprehensive theoretical foundations for player valuation.
- **Hakes, J. & Sauer, R. (2006).** "An Economic Evaluation of the Moneyball Hypothesis." *JEP* 20(3): 173–186. OBP was $\sim 2\times$ as important as slugging for wins but systematically underpriced [ResearchGate](#) 1999–2003; correction visible by 2004 consistent with market learning after *Moneyball*. **Canonical "market inefficiency then correction" empirical case.** Follow-up (Hakes & Sauer 2007, *IJSF*) shows Eye/Power coefficient ratio tripled from 1.67 (1986–93) to 5.36 (2004–06).
- **Romer, D. (2006).** "Do Firms Maximize? Evidence from Professional Football." *JPE* 114(2): 340–365. Dynamic programming on 1998–2000 NFL plays shows teams make statistically significant suboptimal 4th-down decisions: of 3,555 situations where "go for it" was preferred, teams went only 1,426 times. Representative TD-attempt vs. FG choice improves win prob by ~ 3 pp. **Canonical "firms don't maximize" sports test.** [Hockeyanalytics](#)

1B. EMH in sports labor markets

- **Hakes & Sauer (2006), op. cit.** — best-known applied EMH test.
- **Massey, C. & Thaler, R. H. (2013).** "The Loser's Curse: Decision Making and Market Efficiency in the NFL Draft." *Management Science* 59(7): 1479–1495 (NBER WP 11270, 2005). **Most directly relevant paper for this thesis.** Compares market prices of draft picks to surplus value (performance value – rookie salary). Finds top picks systematically overvalued; [Hockeyanalytics +2](#) surplus value actually peaks in late-first / early-second round. [arXiv](#) [ResearchGate](#) Between 1988–96 pick #10 was worth $\sim 60\%$ of pick #1 in trades; by 1997–2004 only $\sim 39\%$. [ResearchGate](#) Attributes to non-regressive prediction, overconfidence, winner's curse, false consensus. [Hockeyanalytics +2](#) Inconsistent with rational expectations. [SSRN +2](#)
- **Brill, R. & Wyner, A. (2024).** "The Loser's Curse and the Critical Role of the Utility Function." arXiv:2411.10400. **Important caveat:** the M-T finding depends on a specific expected-surplus utility; under alternative (risk-seeking / star-seeking) utilities, GM behavior can be rationalized. [arXiv](#) Working paper, not yet peer-reviewed, but a needed nuance.
- **NHL-specific MRP/discrimination work (peer-reviewed):**
 - Jones & Walsh (1988), "Salary Determination in the NHL," *ILR Review* 41(4).
 - Jones, Nadeau & Walsh (1999), "Ethnicity, Productivity and Salary," *Applied Economics* 31(5).
 - Idson, T. & Kahane, L. (2000), "Team Effects on Compensation: An Application to NHL Salary Determination," *Economic Inquiry* 38(2): 345–357. [EconBiz](#)
 - Kahane, L. (2005), "Production Efficiency and Discriminatory Hiring Practices in the NHL: A Stochastic Frontier Approach," *Review of Industrial Organization* 27: 47–71. [Springer](#)

1C. Behavioral economics in sports decision-making

- **Staw, B. & Hoang, H. (1995).** "Sunk Costs in the NBA." *Administrative Science Quarterly* 40(3): 474–494. [Gwern](#) [Semantic Scholar](#) Earlier-drafted

NBA players get more minutes and longer careers than later-picked peers with equivalent productivity. [Gwern](#) Effect $\approx$ 22.8 min/draft position in year 2, 13.8 min in year 5. [Temple](#) **Classic sunk-cost field test.** Leeds, Kowalewski & Suppe (2015) RD design find no robust effect; [ResearchGate](#) Keefer (2015, 2017) and Hinton & Sun (2019, *Empirical Economics*) [Semantic Scholar](#) reconfirm with different measures [Semantic Scholar](#) — **debate is active.**

- **Pope, D. & Schweitzer, M. (2011).** “Is Tiger Woods Loss Averse?” *AER* 101(1): 129–157. [American Economic Association](#) [Clemson](#) Pro golfers make birdie putts $\sim$ 2 pp less often than statistically identical par putts [Traviswhitecommunications](#) — real-world loss aversion under high stakes. Costs top golfers $\sim$ \$1.2M/year. [SSRN](#) Proves experience + stakes don’t debias.
- **Gilovich, Vallone & Tversky (1985),** *Cognitive Psychology* — original “hot hand fallacy.”
- **Miller, J. & Sanjurjo, A. (2018).** “Surprised by the Hot Hand Fallacy? A Truth in the Law of Small Numbers.” *Econometrica* 86(6): 2019–2047. [IDEAS/RePEc](#) [The Econometric Society](#) Discovered a previously unrecognized streak-selection bias in GVT’s 1985 estimator; after correction, **original 76ers data show significant hot-hand evidence** [Wikipedia](#) [SSRN](#) — reversing 30-year consensus. Cite both papers to reflect current state.
- **Camerer & Weber (1999)** *JEBO* — partial confirmation of sunk cost; **Borland, Lee & Macdonald (2011)** replicate in AFL.

Consensus vs divergence: Labor-market inefficiency findings (Scully monopsony, Moneyball OBP mispricing, Massey-Thaler draft overvaluation, Romer risk aversion) form a consistent picture that professional sports markets display persistent, systematic biases — particularly around (a) valuing recent/visible outputs over predictive metrics, (b) overweighting high-profile assets, and (c) risk-averse decision-making under asymmetric career risk for decision-makers (GMs, coaches). This directly motivates the thesis.

2. Hockey analytics — technical benchmarks

2A. Expected Goals (xG) models

Lineage: **Ryder (2004)** foundational “Shot Quality” ([hockeyanalytics.com](#)) → **Krzywicki (2005, 2010)** logistic refinements → **Macdonald (Sloan 2012)** first cited hockey xG [Annual Reviews](#) → **Springs & Toumi (2015)** popularized it [Substack +2](#) → **Perry/Corsica (2016–17)** logistic, deliberately omits shooter talent [Annual Reviews](#) → **Tanner/MoneyPuck (2016)** GBM with flurry adjustment [Evolving-Hockey](#) → **Younggren & Younggren/Evolving-Hockey (2018, rev. 2021)** XGBoost, four strength-specific sub-models.

Input features (typical): shot distance, [NHL](#) shot angle (top-2 features by importance), [Inside The Rink](#) shot type, seconds/distance since last event (proxies rebounds and rush), prior event type/team, strength state, [NHL](#) empty-net flag. [moneypuck](#) [The Hockey Writers](#)

Methodological divergences:

- Algorithm: logistic (Perry, Barlowe) vs. GBM/XGBoost (MoneyPuck, Evolving-Hockey). [Evolving-Hockey](#)
- Shooter talent: Springs yes; Perry/EH/MP explicitly no. [Evolving-Hockey](#)
- Rebound/rush: explicit dummies vs. derived from prior-event distance/time. [NHL](#)
- Strength state: single model vs. separate. [Evolving-Hockey](#)
- Flurry adjustment (sequential rebounds discount): only MoneyPuck. [Evolving-Hockey](#)
- Training era drift: models trained on 2007–18 degrade as shot profile shifts toward quality ([hockey-statistics.com](#), 2022). [GitHub](#)
- **Consequence for thesis:** Evolving-Hockey and MoneyPuck xG can disagree by 5–10 goals per player-season; a trade-valuation thesis should pick one methodology explicitly.

Citations: Younggren & Younggren (2018, updated 2021), [evolving-hockey.com](#) blog; Tanner, MoneyPuck glossary ([moneypuck.com/glossary.htm](#)); Macdonald (2012, MIT Sloan); Ryder (2004, [hockeyanalytics.com](#)).

2B. Goals Saved Above Expected (GSAx)

Formula: **GS_{Ax} = xGA – GA.** [Threads](#) Emerged $\sim$ 2016–2018 as a direct corollary of public xG models; MoneyPuck popularized goalie leaderboards.

Typical single-season ranges (full-season starter, 50+ GP): Elite/Vezina $\sim$ +20 to +30 (Hellebuyck career leader); [LastWordOnSports](#) Good starter +5 to +15; Average $\sim$ –3 to +3; Below $\sim$ –5 to –15; Poor worse than –15. Single-game $>+3$ = “stolen game.” [The Hockey Writers](#)

Limitations: Depends on upstream xG model (cross-source divergence); no pre-shot passing/lateral data in public xG; rebound control not captured; score-state confounding; small samples for backups. [hockeystats.com](#) flagged $\sim$ 332-goal cumulative miscalibration in MoneyPuck’s legacy model through half of 2025–26. [Hockey Stats](#)

2C. Regularized Adjusted Plus-Minus (RAPM)

NBA origin: **Sill (2010, Sloan)** and Engelmann’s xRAPM — ridge (L2) regression shrinks [Medium](#) raw APM standard errors [NBASTuffer](#) driven by teammate/opponent multicollinearity.

NHL: **Macdonald (2011–12)** first APM papers; **Younggren & Younggren (2019, Evolving-Hockey)** currently standard implementation [Hockey Graphs](#) — stint-level weighted ridge [Evolving-Hockey](#) ([glmnet](#) , $\alpha=0$, 10-fold CV) [Hockey Graphs](#) regressing per-60 rates (GF, CF, xGF per 60) [Evolving-Hockey](#) on player dummies, [Evolving-Hockey](#) score state, zone start, home/away, rest; separate regressions by strength state; [Evolving-Hockey](#) mid-season trades

treated as split player observations. [Evolving-Hockey](#) [Evolving-Hockey](#)

Limitations: Coefficients shrunk toward zero, [Hockey Graphs](#) [Hockey Graphs](#) so confidence intervals non-trivial [Medium](#) (Bacon 2021, TopDownHockey); collinearity of frequently-paired linemates (Sedins); [Evolving-Hockey](#) purely public play-by-play — no passing/tracking; linear zone-start/score-state dummies miss non-linear interactions.

2D. Game Score (Luszczyszyn)

Origin: [Luszczyszyn \(July 2016, Hockey Graphs\)](#) [Hockey Graphs](#) — inspired by Bill James & Hollinger. [All About The Jersey](#) Original formula:

$GS = 0.75 \cdot G + 0.7 \cdot A1 + 0.55 \cdot A2 + 0.075 \cdot SOG + 0.05 \cdot BLK + 0.15 \cdot PD - 0.15 \cdot PT + 0.01 \cdot FOW - 0.01 \cdot FOL$ [\[!\[\] \(claude-citation:/icon.png?validation=2BDE8484-FD1E-4125-B7C1-8419B8BAFBA&citation=eyJlbmRjbmRleCI6MTAxMzEsIm1ldGFkYXRhIjp7Im1jb25VcmwiOiJodHRwc2pcLlwd3d3Lmdvb2dsZS5jb21cL3MyXC9mYXZpY29uc29zej02NCZkb21haWw="Substack"\)\]](#) https://shrinkingthegame.substack.com/p/creating-game-score-for-the-pwhl?publication_id=1695808&post_id=155194900&triggerShare=true&isFreemail=true&r=13p19&triedRedirect=true) + 0.05 · CF - 0.05 · CA + 0.15 · GF - 0.15 · GA [All About The Jersey](#) [Substack](#)

Goalie: $GS = -0.75 \cdot GA + 0.1 \cdot SV$. [Blueshirts Breakaway](#) 2019–20 revision replaced Corsi with xG and applied position-specific weighting (D get larger xGF/xGA multipliers to counter scoring bias). [Buhockeystats](#) **GSVA** (Game Score Value Added) is the 3-year weighted rolling Game Score translated to wins [All About The Jersey](#) [Blueshirts Breakaway](#) — core of Luszczyszyn's trade-value and contract-surplus framework at The Athletic. Thresholds: ~0 replacement; F >3 or D >2.5 elite. [All About The Jersey](#)

2E. NHL Equivalency (NHLe) coefficients — comparison table

Formula: $NHLe = (P/GP_sourceLeague) \times 82 \times factor$ [All About The Jersey](#) [All About The Jersey](#)

League	Desjardins (2004)	Vollman (2017)	Perry (2017)	Bacon (2020, network)
KHL	0.77	0.74 The Leaves Nation	0.77	0.772 Towards Data Science Medium
SHL	0.56	0.58	0.56	0.566 Towards Data Science Medium
Liiga	0.54	0.43	0.44	0.441 Towards Data Science Medium
NLA (Swiss)	0.48	0.43	0.46	0.459 Towards Data Science Medium
DEL	—	0.35	0.35	0.352 Towards Data Science Medium
Czech Extraliga	0.52	0.53	0.58	0.583 Towards Data Science Medium
AHL	0.45	0.47	0.39	0.389 Towards Data Science Medium
NCAA — NCHC	—	0.43	—	—
NCAA — Hockey East	—	0.38	—	—
NCAA — Big Ten	—	0.33 The Leaves Nation	—	—
NCAA — ECAC	—	0.23 The Leaves Nation	—	—
NCAA (aggregate)	0.41 The Copper & Blue	—	—	0.194 Towards Data Science Medium
OHL	0.30	0.30	0.25–0.30	0.144 Towards Data Science Medium
WHL	0.28	0.29 The Leaves Nation	0.25	0.141 Towards Data Science Medium
QMJHL	0.24	0.25	0.22	0.113 Towards Data Science Medium
USHL	~0.19	0.22	0.18	0.143 Towards Data Science Medium

Divergence flags: (a) AHL: Perry/Bacon values (~0.39) are notably lower than Desjardins/Vollman (0.45–0.47) because the newer models correct for survivor bias among direct AHL→NHL jumpers. (b) CHL: Vollman's values are roughly 2× Bacon's, driven by multi-path (AHL-transit) vs. direct-jump methodology. (c) Bader/Hockey Prospecting does not publish a fixed coefficient table — uses percentile-ranked production within league/position/age. [Substack](#) Citations: Desjardins (2004, hockeyanalytics.com PDF); Vollman (2017, Hockey Abstract); [Substack](#) Bader (2019, hockeyprospecting.com); Bacon (2020, Towards Data Science); Turtoro (2019/2020, network-based NHLe). [Substack](#) [Hockey Stats](#)

2F. Aging curves

Consensus: Peak age ≈ 24–26 for forwards, modestly later/flatter for defensemen. Survivorship-corrected methods (McCurdy 2022) preserve this finding; legacy delta-plus-minus methods (Brander, Egan, Yeung 2014 JQAS) push peak to 27–28 (F) / 28–29 (D) [Simon Fraser University](#) [ResearchGate](#) but are criticized for metric choice. Components age differently: skating/quickness earliest; hockey-IQ/positioning/faceoffs latest.

Key works: Tulsky (2011–13, Broad Street Hockey / NHL Numbers); Desjardins (2009, Arctic Ice Hockey); Hockey Graphs aging curve series (2017, two

parts with Lichtman “phantom” survivorship correction); McCurdy (2022, “Age and Decay,” HockeyViz); [Hockey Graphs](#) Brander, Egan, Yeung (2014, JQAS); [ResearchGate](#) Swartz et al. (2024, JQAS, FPCA method); Schuckers, Lopez & Macdonald (2022 preprint, imputation-based). [Hockeyviz](#) [Carnegie Mellon University](#)

Implication for thesis: Contracts extending past age 30 routinely under-deliver — well-documented market inefficiency. Survivorship bias ensures observed old-player contracts skew upward.

2G. Quality of Competition (QoC) and zone starts

QoC metrics: TOI% QoC (Desjardins), CF% QoC, xG QoC, Woodmoney/PuckIQ bins. Tulsky (2012, NHL Numbers) showed the season-long range of QoC faced is **narrow (~±2 Corsi/60)**. [Hockey Graphs](#) Hohl (2015, Hockey Graphs): std dev of season CorC% ~0.3–0.4 pp; teammate CorT% ~2.5 pp [Hockey Graphs](#) — **teammates matter ~5–6x more than competition.** [Hockey Graphs](#) QoC washes out over full seasons; [Hockey Graphs](#) relevant for short-sample and junior/NCAA levels.

Zone starts: Tulsky found ~0.31 Fenwick per o-zone start — about half of [Hockey Graphs](#) Desjardins’s original estimate. [Hockey Graphs](#) Most players’ OZS% between 40–60%. [Hockey Graphs](#) Most shifts (~70–80%) start on-the-fly, not at faceoffs, [The Hockey Writers](#) bounding the impact. Modern RAPM/WAR models include ZS as a linear dummy.

Thesis implication: Markets that price players on “tough minutes” narratives likely misprice — QoC signal is small; teammate quality is the bigger unobserved confound.

3. NHL contract & cap mechanics

3A. RFA vs UFA rights

- **Group 2 RFA:** post-ELC, <27 years old, <7 accrued seasons. [Sound Of Hockey](#) Only QO'd Group 2s may sign offer sheets. [Sound Of Hockey](#)
- **Group 6 UFA:** 25+ at June 30, 3+ pro seasons, expiring contract, [Puckpedia](#) fewer than 80 NHL games (28 + 30 min for goalies). [Puckpedia](#)
- **Group 3 UFA:** 27+ OR 7+ accrued seasons.
- **QO tiers** (prior salary): <\$660K → 110%; \$660K–\$1.0M → 105%; >\$1.0M (post-July 2020) → lesser of 100% prior [Wikipedia](#) or 120% of cap hit. [Puckpedia](#) QO must be one-way if player had 180+ GP last 3 seasons and 60+ GP prior season and did not clear waivers. [Puckpedia](#) Deadline: later of Monday after Entry Draft or June 25 5 PM ET; player must sign by Dec 1. [Puckpedia](#) [Wikipedia](#)
- **Arbitration eligibility by first-contract age:** 18–20 → 4 yrs pro; 21 → 3; 22–23 → 2; 24+ → 1. [NHL](#) Team-elected walk-away threshold set from league-avg salary: ~\$4.538M (2021), [Wikipedia](#) ~\$4.58M (2023), ~\$4.74M (2024), [Pro Hockey Rumors](#) ~\$4.85M (2025). [Puckpedia](#)

Offer sheet compensation tiers 2025-26 (source: PuckPedia; AAV = total comp / min(years, 5)):

AAV (2025-26)	Compensation
<\$1.54M	None
\$1.54M–\$2.34M	3rd
\$2.34M–\$4.68M	2nd
\$4.68M–\$7.02M	1st + 3rd
\$7.02M–\$9.36M	1st + 2nd + 3rd
\$9.36M–\$11.7M	2x1st + 2nd + 3rd
>\$11.7M	4x1st

Recent example: St. Louis signed Philip Broberg [CanucksArmy](#) (2yr, \$4.58M) and Dylan Holloway (2yr) as RFA offer sheets from Edmonton in summer 2024 [Puckpedia](#) — first successful offer sheets since Kotkaniemi (CAR) 2021. [NHL](#) Citations: PuckPedia; The Fourth Period; Pro Hockey Rumors.

3B. Entry Level Contract (ELC) structure

Length by signing age (Sept 15): 18–21 → 3-yr; 22–23 → 2-yr; 24 → 1-yr; [Pro Football Network](#) 25+ → no ELC. All ELCs are two-way. [VIEW FROM CENTER ICE](#) Maximum base salary: \$925K (2019–21), \$950K (2022–23), **\$975K (2024–25)**. [Pro Football Network](#) Signing bonus ≤10% of salary; [Pro Football Network](#) AHL max ~\$82.5K. [VIEW FROM CENTER ICE](#) Schedule A (common, tied to GP + statistical/award thresholds): ~\$212.5–250K per category, up to 4 categories, \$850K–\$1M cap. [SportsOrca](#) Schedule B (rare, major awards): up to \$2.5M for recent ELCs. [SportsOrca](#) Combined elite ELC cap hit (Matthews/Bedard) ≈ \$3.775M. **Slide rule:** 18–19-year-olds with <10 NHL GP slide the ELC one year; [Pro Football Network](#) 10th NHL game burns a year. [SportsOrca](#)

Why ELCs create surplus value: Fixed low cap hit (~\$975K base) for elite young production is the single largest source of surplus value driving NHL trade markets — underpins the entire “cost-controlled asset” thesis.

3C. Salary retention (CBA Art. 50.5(e))

- ≤50% of salary and cap hit (same %, fixed over contract life). [The Hockey Writers](#)
- Max 3 retained contracts per team at any time. [Puckpedia](#)
- Aggregate retention cap: 15% of upper limit (≈\$14.3M at \$95.5M cap).
- A contract can be involved in max 2 retention transactions [The Hockey Writers](#) (“double retention” → 25% effective).
- Trade-back restriction: retaining team cannot reacquire player for 1 calendar year. [Puckpedia](#)
- **2025-26 amendment:** cannot retain on a contract already retained within 75 regular-season days — effectively ends same-transaction double retention; changes future third-party broker strategy.

Notable examples: Mikko Rantanen three-team deal (COL→CAR via CHI, Jan 2025 — Chicago retained 50% of \$9.25M = \$4.625M); [Bleacher Report](#) Seth Jones to Florida (Chicago retained \$2.5M × 5 yrs — largest retained cap hit in history). Citations: PuckPedia; The Hockey Writers; Offside News.

3D. No-Trade / No-Movement clauses

Eligibility: Group 3 UFA status — 27+ years old OR 7+ accrued seasons. **The original prompt’s “7+ accrued AND 4+ years pro” is inaccurate — that formula is arbitration eligibility for age-18–20 first signees.** Flag for the thesis.

- **NTC:** no trade without consent; can still be waived/sent to minors.
- **NMC:** no trade, waive, expansion exposure, or minor-league assignment without consent. Must be protected in expansion drafts. Can be bought out; bypasses standard waiver process.
- **Modified NTC / Limited NTC:** player submits N-team allow-list or block-list (commonly 8–20 teams).

Prevalence: Approximately 100–130 players (10–15% of league) carry some form of trade protection in 2024-25 (Pro Hockey Rumors / The Fourth Period annual lists).

Published discount: No peer-reviewed quantification exists. Practitioner estimates (Evolving-Hockey contract model, PuckPedia capology pieces) suggest players extract ~3–8% additional AAV or ~1 extra year for trade protection. **This is a genuine gap and a viable thesis contribution** — measure the residual in an AAV-vs-projected-value model conditioning on NTC/NMC presence.

3E. Salary cap history and projections

Season	Ceiling	Notes
2005-06	\$39.0M	First cap year
2013-14	\$64.3M	
2019-20	\$81.5M	COVID
2020-21 to 2021-22	\$81.5M	Flat (COVID)
2022-23	\$82.5M	
2023-24	\$83.5M	
2024-25	\$88.0M	Post-COVID
2025-26	\$95.5M	+\$7.5M
2026-27 (proj.)	\$104.0M	NHL-NHLPA Jan 2025 agreement
2027-28 (proj.)	\$113.5M	

Discrepancy flag: Friedman (Sportsnet, Oct 2025) reported the 2026-27 figure could adjust upward to ~\$107M on stronger HRR (Rogers 11-yr/\$11B CAD Canadian media deal, April 2026). Salary floor = ceiling × (0.85/1.15) — 2025-26 floor \$70.6M.

4. Draft pick valuation literature

4A. Michael Schuckers

- **Schuckers (2011)** “What’s an NHL Draft Pick Worth?” SLU working paper. LOESS on 1988–97 drafts. Pick #1 = 917 career GP; #10 = 565; #50 = 185; #100 = 107; #210 = 51. **Ratio ~18:1**, far less extreme than Jimmy Johnson’s ~500:1 NFL chart. 200+ GP probability: #1 ≈ 98%; #210 ≈ 8%. Hosted at [myslu.stlawu.edu](#).
- **Schuckers & Argeris (2015)** “You Can Beat the Market.” *Journal of Sports Analytics* (arXiv 1411.5754). Uses first 7 years of TOI as response; internal

scouting worth \$1.7–5.2M/team/year vs. NHL Central Scouting.

- **Schuckers (2016)** updated chart (2003–2008 drafts, TOI, monotone smoothing) — steeper curve; pick #1 normalized to 1000.
- **Moreau, Perera & Swartz (2025)**. “Valuation of NHL draft picks using functional data analysis.” *Journal of Sports Analytics* — updated FDA-based curve.

4B. Eric Tulsky (2013, Broad Street Hockey)

Market-based (trade-revealed) pick values from 46 pick-only trades 2006–2012. Pick #1 = 100 normalized. Selected values: #2 = 69.9; #3 = 59.4; #5 = 49.2; #10 = 36.5; #20 = 22.4; #30 = 14.2; #45 = 7.2; #60 = 3.8; #90 = 1.1; #210 = 0.10. **Tulsky’s market curve is steeper than Schuckers’s performance curve** — analogous to Massey-Thaler finding in NFL: GMs overvalue top picks relative to actual surplus value. Tulsky named Carolina Hurricanes GM in June 2024.

4C. NFL comparisons

- **Jimmy Johnson chart (1991, Cowboys)**: Pick #1 = 3,000 points, ~500:1 ratio.
- **Massey & Thaler (2013)**: surplus value non-monotonic, peaks mid-first round; 1988–96 #10 was 60% of #1 in trades, 1997–04 only 39%.
- **NHL vs NFL**: NHL curves flatter than NFL Johnson but steeper than NFL surplus-value curves. Drivers: (a) longer development curve; (b) higher NHL bust rate — only ~40% of #25 overall picks reach 200 GP; (c) NHL rookie scale is essentially flat (\$975K) regardless of slot, so NHL surplus-value curve tracks performance more closely.

4D. Probabilistic weighting for future/conditional picks

No published standard. Industry best practice (Hockey Prospecting, DobberProspects, The Athletic): (i) distribution over possible final slot from current-season standings, (ii) time-discount factor (10–20%/year typical in practitioner analyses), (iii) lottery odds (NHL 2022 reform: top 11 eligible; max 10-slot jump; max 2 wins per 5-year period), (iv) conditional clauses folded into a truncated expectation. Tulsky’s 2013 paper **excluded future/unresolved picks** explicitly, acknowledging the complexity.

4E. Bust rates by round

- **Schuckers (2011), % reaching NHL (≥ 1 GP)**: R1 = 91.7%; R2 = 65.0%; R3 = 56.7%; R4 = 38.0%; R5 = 32.3%; R6 = 27.3%; R7 = 21.0%.
- **HFBords (1979–2001, 250+ GP)**: R1 = 61%; R2 = 30%; R3 = 24%; R4 = 15%; R5 = 12%; R6 = 12%; R7 = 7.5%.
- **DobberProspects (2003–2012, 100+ GP)**: R1 = 74%; R2 = 34%; sharp drop at pick 25–31. All top-5 picks reached 100+ GP.
- **Neutral Zone (2018, 200+ GP)**: pick #25 $\approx$ 40%; R3–R7 $\approx$ <10%.
- **Position** (Schuckers, 200+ GP): F 25%, D 21%, G 19%.

Methodological caveat: Studies use different GP thresholds (50/100/200/250) and different draft eras — numbers are not directly comparable. Post-2005 cap era shows higher success rates, likely due to earlier promotion driven by ELC cap value.

5. Prospect projection literature

5A. Bayesian / probabilistic approaches

No fully Bayesian NHL prospect system in the public literature, but logistic-regression probability frameworks (Bader, Bacon) effectively deliver probabilistic outputs.

5B. Byron Bader — Hockey Prospecting

Logistic regression on NHL-e-transformed scoring rates; parallel NHLer and Star probability models at each developmental stage (D-1 through D+3 skaters; D-1 through D+5 goalies). Trained on 5,000+ skaters drafted R1–R7, 1990–present, plus 1980s stars, and ~700 goalies.

Thresholds: NHLer = 200+ career GP (100+ for goalies); Star (F) = 0.70+ career PPG (~60+ pts/season); Star (D) = 0.45+ career PPG. Only ~3.5–4% of all drafted skaters historically hit “Star.”

Published findings:

- 77% of eventual NHLers hit a 30+ NHL season in junior; >90% of “impact point-producers” (0.6+ career PPG) had ≥ 1 such season.
- Of 846 R2–R7 prospects (2000–2013) with $\leq 30\%$ modeled NHLer probability, only 86 (10%) reached 200 GP; 31% of those played significantly only for a different team.
- Of top-35 Star-probability skaters 2013–16, 28 (80%) became Stars.
- Birth-month effect: late-born (Q3/Q4) R1 skaters succeed ~10% more often than early-born R1 peers.

Citations: Bader (2019, 2021), hockeyprospecting.com; San Jose Hockey Now interview (2021).

5C. Namita Nandakumar

Currently Director of R&D, Seattle Kraken (prior: Philadelphia Eagles, Wharton 2018). Key works:

- **“Exploiting Variance in the NHL Draft”** (Hockey Graphs, 2017) — argues GMs should prioritize high-variance (boom/bust) prospects over low-ceiling floor picks.
- **Wharton thesis (2017)**, “What Does It Mean To Draft Perfectly? An Evaluation of Draft Strategy in the NHL” — defines “Perfect Draft Value” via backwards best-player-available; finds no team sustained drafting edge 2000–09; correlation $r = 0.45$ between perfect and actual value. Conclusion: “draft early and often.” Repository: repository.upenn.edu/joseph_wharton_scholars/26.
- **Nandakumar & Jensen (2019)**, “Historical Perspectives and Current Directions in Hockey Analytics,” *Annual Review of Statistics and Its Application* 6: 19–36. DOI 10.1146/annurev-statistics-030718-105202 — **primary peer-reviewed survey of hockey analytics**; extended NHL draft section.
- RITSAC 2017 “Survival Analysis of NHL Prospect Timelines”; RITSAC 2019 “NHL Player Size Impacts Performance.”

5D. Cohort models (pGPS / PCS / SEAL / Network NHLe)

- **Weissbock & Lawrence (2014–15), PCS** — Prospect Cohort Success. Match on age, height, PPG; outcome = share reaching 200 NHL GP. Both hired by **Florida Panthers in June 2016**; consulting contracts not renewed in 2021; Weissbock later reportedly returned to Ottawa Senators.
- **Jeremy Davis (2016–present), pGPS** (Canucks Army, later NextGen Hockey) — expected success % + expected points/82 using cohort similarity; later expanded with SEAL adjustments and production thresholds (not just 200 GP).
- **Hohl & Jessop (2016), SEAL-adjusted scoring** — NHLe plus age and situational (5v5 vs PP) adjustments.
- **Turtoro (2020), Network NHLe** — graph-theoretic, uses multi-league transitions (not just league→NHL) to impute translation factors for leagues with few direct NHL graduates.
- **Bacon (2021)** logistic regression onto WAR-based NHLer/Star labels, D+8 bust cutoff. hockeystats.com/methodology/nhle.
- **Wylie et al. (2020)**, Elo-paired league strength (Hockey Graphs, “Which League is Best?”).

“DEV” (Draft Expected Value): No single definitive public source with that name. Likely references Nandakumar’s “Perfect Draft Value” or Luszczyszyn’s GSVA draft curve, or may be an internal term. **Recommend the thesis clarify.**

6. Systematic biases evidence in NHL trades

6A. Size / grit / veteran bias

- **Dubas (Sloan 2015)** “How Analytics Has Limited the Impact of Cognitive Bias on Personnel Decisions.” Identified confirmation, recency, information, sample-size, and simplicity biases. Specifically flagged trade deadline and training camp as bias-vulnerable periods — e.g., GMs overweighting a single recent viewing.
- **Hohl (Hockey Graphs, 2017; The Five Hohl 2023/2026)** — repeated finding that size/hitting has no meaningful correlation with playoff success across 2014–2016 playoffs.
- **Canonical “grit premium” example:** Tanner Jeannot trade (Feb 2023, TB → NSH sent 2024 1st, 2nd, 3rd, 4th, 5th + Cal Foote for Jeannot — a bottom-six hitter with one 24-goal season).

6B. Trade deadline premium / winner’s curse

- **Luszczyszyn, The Athletic GSVA trade-value framework (2020, updated annually)** — consistently grades deadline rental acquisitions as negative surplus value for buyers (paywalled; referenced in NHL.com companion pieces and fan-site summaries).
- **Thomas / Sound of Hockey (2022)** — trade-based pick value chart benchmarked against Luszczyszyn’s GSVA: “near the top of the draft, trade value has exceeded projected GSVA of players actually selected” — direct evidence of systematic GM overvaluation of top picks.
- **Schuckers (2011) + Moreau et al. (2025)** — late-first-round picks have only ~15% probability of producing top-6F / top-4D, making the standard rental = late 1st calculus a negative-EV proposition in expectation.
- **Tulsky vs Chuckers gap** (Section 4) is the single cleanest existing empirical signature of NHL “loser’s curse.”

6C. Rental player pick costs — case studies

- **Rick Nash** (NYR → BOS, Feb 25 2018; NYR retained 50%): 2018 1st, Ryan Spooner, Matt Beleskey, Ryan Lindgren, 2019 7th. Nash: 18G/28 pts in 60 GP. Only one 1st-round pick had moved across both prior deadlines combined.
- **Mark Stone** (OTT → VGK, Feb 25 2019; 8-yr/\$76M extension same day): Erik Brännström, Oscar Lindberg, 2020 2nd.
- **Jakob Chychrun** (ARI → OTT, March 1 2023; 2+ yrs at \$4.6M): 2023 conditional 1st (became #12 — Daniil But), 2024 2nd, 2026 2nd. Luszczyszyn valued Chychrun at ~\$10.4M market vs \$4.6M cap → \$5.8M/yr surplus.
- **Tomáš Hertl** (SJS → VGK, March 8 2024; SJ retained 17%, 6 yrs at \$8.137M): David Edstrom (2023 1st #32), 2025 1st; VGK received 2025 and 2027

3rd back.

- **Jake Guentzel** (PIT → CAR, March 8 2024; PIT retained 25%, pure rental): Michael Bunting, Ville Koivunen (2021 2nd, #51), Vasily Ponomarev (2020 2nd, #53), Cruz Lucius (2022 4th), conditional 2024 2nd (convert to 1st if Cup Final — didn't), conditional 2024 5th. Guentzel walked UFA to TB for only a 3rd-rounder in sign-and-trade. The Hockey News graded CAR as “losing big.”

Sources: NHL.com trade records; PuckPedia trade entries; Elite Prospects; The Hockey News retrospectives.

6D. Post-trade regression to mean

- **Tulsky, Arctic Ice Hockey (~2013–14)** — foundational PDO regression study. PDO through first 1,000 shots regresses **87% to the mean** over the rest of the season. A team at 1030 PDO through 1000 shots expects 1004 rest-of-season. Direct quantitative basis for “hot players regress post-trade.”
- **Wendorf (Hockey Graphs, 2014)** — team PDO converges to 975–1025 range full-season.
- **Luszczyszyn (Leafs Nation counterpoint, 2023)** — acknowledges regression even while arguing GMs can lean into recency for 20-game rentals.

6E. Anchoring on comparable contracts

- **Goldman & Murphy (ISOLHAC 2020)** — GMs explicitly “draft lists of comparable contracts”; arbitration rulings typically land near the midpoint of team/player filings (“awards fall very close to the simple average of the ask and offer” — PPP Leafs). This midpoint-anchoring mechanism institutionalizes comparable anchoring into actual rulings.
- **PuckPedia 2024/2025 arbitration thresholds:** comparables admissible; advanced-stat charts (RAPM) are not — forcing anchoring on traditional counting stats.
- **Matt Cane**, “Time Value of Money and Player Valuation” — foundational industry framework; cited by Idriss (Hockey Graphs 2021) and Goldman for NPV-style contract valuation.

7. Historical trade datasets & case studies

7A. Datasets comparison

Source	Coverage	Key fields	Access
PuckPedia (puckpedia.com)	Current + historical, real-time	Full trade details, retained salary, contract terms, 3-team mechanics	Free; also PuckGM simulator
CapFriendly	2016–2024 (defunct; acquired by WSH 2024)	Historical gold-standard — contracts, trades, Armchair-GM	Archive only via Wayback Machine
CapWages (capwages.com)	Post-CapFriendly replacement	Similar UI	Free
Spotrac NHL (spotrac.com/nhl/transactions/trade/)	Current + historical	Retained salary, cap breakdowns	Free
NHL Trade Tracker	Historical	Indexed by team/player/GM; notable-trades sections	Free
Hockey-Reference	Full historical	Transactions + Stathead search	Free basic; Stathead subscription
Elite Prospects	Broad international	Transaction pages, trade trees, prospect data	Free + paid
HockeyDB	Deep historical	Machine-readable CSV via github.com/rippinobr/hockeydb	Free
Evolving-Hockey	2007-08+	GAR/xGAR/RAPM, contract projections	Subscription (\$5/mo)
MoneyPuck	Full PBP era	xG, shot-level CSV exports	Free
Natural Stat Trick	Full PBP era	Advanced stats by year	Free

Dataset recommendation for the thesis backtest: PuckPedia (current + recent historical trades with retained salary breakdowns) + Hockey-Reference/Stathead (comprehensive historical baseline) + Evolving-Hockey (player value metrics). Supplement with Wayback Machine CapFriendly archives for pre-2024 contract details. Retained-salary is the critical field frequently missing from older datasets.

7B. Notable case studies

Trade	Date	Consensus retrospective

Taylor Hall / Adam Larsson	Jun 29, 2016	NJ won talent exchange (Hall Hart 2018); Larsson reliable top-4 but never #1.
P.K. Subban / Shea Weber	Jun 29, 2016	Subban led NSH to 2017 Final; Weber led MTL to 2021 Final. Rare mutual impact.
Seth Jones / Ryan Johansen	Jan 6, 2016	One-for-one swap of #4 overalls. Rare true win-win.
Erik Karlsson to SJS	Sep 13, 2018	OTT got Tierney, Balcers, DeMelo, Norris, unprotected 2020 1st (became Stützle #3), 2021 2nd. OTT winner even before Stützle emerged.
Erik Karlsson to PIT	Aug 2023	3-team; SJS got PIT 2024 1st (top-10 prot.), Granlund, Hoffman, Rutta; retained 13% (\$1.5M/\$11.5M). Mixed grade.
Jack Eichel to VGK	Nov 4, 2021	BUF got Tuch, Krebs, 2022 1st, 2023 2nd. Eichel won 2023 Cup; 94 pts in 2024-25; 8yr/\$13.5M ext Oct 2025. VGK clearly got the star.
Tkachuk / Huberdeau + Weegar + 1st	Jul 22, 2022	Tkachuk → 3 Finals, 2 Cups. Huberdeau's 8yr/\$84M extension became "worst contract in NHL" (Luszczyszyn). Universally lopsided FLA win.
Ryan O'Reilly to STL	Jul 1, 2018	STL gave Tage Thompson, Berglund, Sobotka, 2019 1st, 2021 2nd. O'Reilly won Conn Smythe + Selke + Cup 2019. Thompson's 47G/94pts 2022-23 rebalanced grade.
Mark Stone to VGK	Feb 25, 2019	OTT got Brännström, Lindberg, 2020 2nd. Stone captain, 2023 Cup. Major VGK win.
Phil Kessel to PIT	Jul 1, 2015	TOR retained \$1.2M/yr; back-to-back 2016/17 Cups for PIT.

7C. Academic papers backtesting NHL trade valuations

- **Schuckers & Argeris (2015)**, *Journal of Sports Analytics* — scouting ROI and pick valuation (adjacent; not direct trade backtest).
- **Moreau, Perera & Swartz (2025)**, *JQAS* — updated FDA pick valuation.
- **Nandakumar & Jensen (2019)**, *Annual Review* — survey.
- **No single comprehensive academic paper has backtested NHL trade valuations systematically.** This is the thesis's contribution gap. NHL betting-market efficiency work exists (Luxton/Spizman/Moore LMU 2022; Traugutt UNCO dissertation on NHL totals) but is unrelated to trade valuations.
- **Mongeon (Brock/Ottawa)** — NHL labor economics and discrimination; adjacent.
- Search MIT Sloan 2013–2024 proceedings — most hockey papers are goaltending (Bayesian Spatial), THoR, or xG; trade-valuation specifically is underexplored.

7D. Aggregate retrospectives

- **Luszczyszyn (The Athletic)** annual trade-grade articles and "NHL Trade Value Chart"; GSVA-based, paywalled but most widely cited.
- **Sean Tierney** trade flow visualizations.
- **Shayna Goldman** trade analyses at The Athletic.
- **Corey Pronman, Scott Wheeler** prospect-centric trade coverage.

8. NPV & discount rate justification

8A. Published practice in sports-asset valuation

- **Matt Cane, "Time Value of Money and Player Valuation"** — foundational practitioner framework applying NPV logic to NHL contracts. Establishes the \$/win-to-NPV translation (cited by Goldman, Idriss, Luszczyszyn).
- **Dom Luszczyszyn GSVA contract-surplus model** — explicitly discounts future-year projected value; surplus = sum of (projected \$value – actual AAV) across contract years, with age decay and implicit time discount embedded.
- **Evolving-Hockey contract projection model (evolving-hockey.com)** — logistic/regression model generating AAV and term predictions; their contract-market model backs out an implicit discount rate consistent with 5–8% annual discount on future production.
- **Schuckers (2011, 2016)** pick value charts — implicit time discount via response-metric choice (career GP / 7-year TOI).

8B. Corporate finance analogues

Standard practice in sports-asset valuation borrows from: (i) **WACC-style discounting** with the discount rate reflecting systematic (injury, performance) +

idiosyncratic (organizational) risk; (ii) **real options valuation** — a prospect or conditional pick is an option on future production (Black-Scholes analogues exist in academic MLB/NFL prospect work); (iii) **DCF with terminal value truncation** at UFA age or contract end.

8C. Justification for differential discount rates

The thesis's proposed 8–10% player rate vs 10–15% pick rate is defensible against standard corporate finance logic:

- **Players under contract (8–10%):** Cash-flow analogue with known term and \$ commitment; primary risks are injury (~5% season-ending injury rate per player-year), underperformance/aging, and team context change. Equity-risk-premium analogues (~6–7%) plus injury premium (~2–3%) justify 8–10%. Matches implied rates in Evolving-Hockey's contract model.
- **Draft picks (10–15%):** Higher uncertainty — the pick is an option on an unknown player (not yet drafted, or drafted prospect with development uncertainty). Bust rates by round (Schuckers, Bader) directly motivate a higher rate: if a R2 pick has ~30% probability of reaching 200 GP, the expected-value uncertainty is far greater than a known NHL player. Additional premia for development timeline (3–5 years typical), organizational uncertainty (coaching/development system variance), and CBA risk (future cap changes).
- **Real options framing:** A draft pick or prospect is a deferred call option on NHL production — higher time value of money and higher vega (option value of variance) both push the effective discount rate upward. Nandakumar's "Exploiting Variance" (2017) provides the hockey-specific theoretical rationale for treating high-variance prospects as valuable options.

Empirical calibration suggestion: The thesis can calibrate player and pick discount rates by solving for the rates that best reconcile Tulskey's market pick curve with Schuckers's performance curve — the implied market discount rate would itself be a publishable finding.

Key flags and caveats for the author

1. **NTC/NMC eligibility in the original prompt ("7+ accrued AND 4+ years pro") is incorrect** — correct CBA language is Group 3 UFA status (27+ OR 7+ accrued seasons). The "4 years pro" refers to arbitration eligibility for age-18–20 first signees.
2. **2026-27 cap projection has two values:** \$104M (formal NHL-NHLPA Jan 2025 agreement) vs. potential ~\$107M (Friedman/Sportsnet Oct 2025, based on Rogers \$11B CAD deal). Use \$104M as baseline; flag upside.
3. **2025-26 retention amendment** (75-day restriction) meaningfully changes the third-party broker strategy for trades post-2025 — any historical analysis bridging pre/post should flag this regime change.
4. **Tulsky vs Schuckers curve gap** is the clearest existing empirical signature of NHL "loser's curse" — should be the central empirical anchor the thesis builds upon.
5. **Massey-Thaler has been contested** by Brill & Wyner (2024, arXiv, not yet peer-reviewed) — the thesis should acknowledge this utility-function critique while noting it remains unpublished.
6. **NHLe coefficients disagree ~2× across methodologies** for AHL and CHL — the thesis must pick a methodology (recommend Perry/Bacon multi-path for survivor-bias correction) and justify it.
7. **Luszczyszyn/Goldman/Tierney work is paywalled** on The Athletic — specific GSVA trade-grade values cannot be fully quoted from primary source; use companion NHL.com and secondary references.
8. **CapFriendly's historical trade database is no longer directly accessible** post-2024 acquisition; PuckPedia has reconstructed most but gaps exist. Wayback Machine is the backstop.
9. **"DEV" (Draft Expected Value) is not a canonical public acronym** — likely a reference to Nandakumar's "Perfect Draft Value" or Luszczyszyn's GSVA draft curve. Clarify with advisor.
10. **Genuine gap in the literature (potential thesis contribution):** no published quantitative estimate of the NTC/NMC contract discount; no systematic academic backtest of NHL trade valuations. These are defensible original contributions.